

Wanderless Laos — Final Group Assignment Report

MGMT 655: Machine Learning for Decision Making | Professor Jack Hong | Singapore Management University MBA Programme

Repository: <https://github.com/joshgd1/wanderless-tourism> **Note:** Professor Hong has been separately granted repository access. Access may be removed after grading as instructed.

Page 1 — Executive Launch Brief

Wanderless Laos: AI/ML Decision-Support for Local Guide Matching in Luang Prabang

Wanderless Laos is a pilot-ready AI/ML decision-support prototype for compatibility-based local guide matching in Luang Prabang, Laos. Instead of acting as another generic itinerary generator, it helps tourists and travel operators decide which local guide, group, and itinerary best fit a traveller's language, budget, pace, interests, safety comfort, and travel style. The prototype uses synthetic but structured data to demonstrate explainable guide matching, segmentation, itinerary logic, safety/trust indicators, and simulated booking. It is not a production deployment, but it is suitable for a controlled pilot with local operators or boutique hotel partners.

Intended Readers

Reader	What This Report Helps Them Decide
Business manager approving launch	Whether Wanderless Laos is ready for a controlled pilot and what risks must be managed
User of the app	How the app helps a tourist choose a compatible guide and understand the recommendation
Fellow developer taking over the app	How the repository is structured, how the app runs, and where the matching logic is documented

Key Dimensions

Dimension	Detail
Core Function	Tourist–Guide compatibility matching + itinerary planning in Luang Prabang
ML Engine	Hybrid recommender: cosine similarity + TruncatedSVD collaborative filtering
Group Formation	K-Means clustering + DBSCAN outlier detection
Itinerary Optimiser	Greedy construction + 2-opt local search
Safety/Trust Scoring	Rule-based decision-support prototype — not a safety guarantee
Backend	FastAPI + SQLite + JWT authentication
Frontend	Flutter cross-platform mobile app

Dimension	Detail
Data	400 synthetic tourists, 60 synthetic guides, 600 synthetic ratings
Tests	14/14 pytest tests passing
Booking / Payment	Fully simulated — no real money moves

Page 2 — Business Problem and Market Need

The Problem: Tourists Cannot Easily Find the Right Local Guide

Tourists visiting Luang Prabang face a specific decision problem: given dozens of local guides with different specialties, languages, and styles, how does a visitor choose someone trustworthy, compatible, and well-suited to their preferences before committing to a multi-day itinerary?

Existing solutions fail on three dimensions:

1. **Information asymmetry.** Reviews on generic platforms are sparse and unverifiable. There is no structured authenticity signal for Luang Prabang's local guide ecosystem.
2. **Itinerary complexity.** Planning a multi-stop trip — balancing opening hours, travel times, meals, energy, and cultural interests — is a combinatorial optimisation problem travellers solve sub-optimally by hand.
3. **Group formation friction.** Solo travellers who want to share a guide's cost have no structured mechanism to find compatible peers.

Why Luang Prabang as a Beachhead

Luang Prabang is a UNESCO World Heritage town with a concentrated, high-value local guide ecosystem: temple ceremonies, Mekong boat trips, Kuang Si waterfalls, night markets, and alms-giving rituals. The tourist profile (slow travel, cultural immersion, small groups) is well-suited to compatibility-based matching, and the defined geographic scope makes it ideal for a data-constrained prototype.

Why Machine Learning

Traditional platforms use keyword search and star ratings. Wanderless Laos upgrades each core decision with ML:

- **Guide matching** uses a hybrid recommender combining content-based preference alignment and collaborative rating patterns — producing explainable compatibility scores, not search results.
- **Group formation** uses K-Means + DBSCAN clustering to surface compatible solo travellers automatically.
- **Itinerary planning** uses greedy construction + 2-opt local search to produce routed day-by-day schedules.

Stakeholder Table

Stakeholder	Pain Point	Value from Wanderless Laos
Tourist	Cannot assess which guide fits preferences and safety comfort	Explainable ranked recommendation with trust indicator
Local guide	Poor-fit bookings waste time and generate bad reviews	Better-matched tourists; clearer profile signals
Hotel / operator	Manual guide matching is inconsistent and unscalable	Data-supported recommendation layer
Destination partner	Visitor experience quality varies with guide	Better trust signals and satisfaction tracking

Page 3 — Product Experience for Users

User Journey: Emma the Solo Traveller in Luang Prabang

Emma is a 30-year-old UX designer from London, travelling solo to Luang Prabang for 4 days. She wants an authentic cultural and food-focused experience but does not know which local guide to choose or how to plan a safe, well-paced itinerary.

Emma's Journey on Wanderless Laos:

Step	User Action	App Output	Decision Supported
1	Enters preferences (food, culture, adventure; pace; budget; language; style)	Structured traveller profile stored in database	What kind of trip is needed
2	Reviews guide list for Luang Prabang	Ranked guide list scored by compatibility	Which guides are most compatible
3	Reads match explanation for each guide	Factor breakdown: preference alignment, collaborative rating, destination match	Why this guide is recommended
4	Checks safety/trust indicator and itinerary	Composite safety/trust score (0–100) + optimised itinerary with travel times	Whether the guide and plan feel suitable
5	Simulates booking and receives confirmation	Booking confirmation with simulated price quote and wallet debit	Whether to proceed with the plan

The booking flow is simulated. The safety/trust indicator supports user judgement but does not guarantee safety.

What the Recommender Actually Does

When Emma requests recommendations, the system: (1) builds a numerical preference vector from her profile; (2) scores each Luang Prabang guide using cosine similarity against guide expertise vectors

(content-based, 45%); (3) looks up collaborative rating patterns via a TruncatedSVD model (collaborative, 45%); (4) applies a destination boost for Luang Prabang-verified guides (10%). The result is an explainable composite score — the business manager, tourist, and developer can each trace which factor drove a guide's ranking.

Page 4 — Business Model, Launch Case, and Final Recommendation

Business Model: B2B2C Platform for Luang Prabang

Wanderless Laos operates as a two-sided marketplace:

- **Supply side (Guides):** Local Luang Prabang guides create profiles with specialty, language, license, and availability.
- **Demand side (Tourists):** Solo and small-group travellers discover guides, view compatibility scores, and book (simulated).
- **Platform (Operators / Hotels):** Boutique hotel partners can integrate the recommendation layer.

Revenue Model: Commission on simulated bookings (15% platform fee, configurable). All bookings are currently simulated — no real payments occur.

90-Day Pilot Plan

Phase	Timeline	Focus	Milestone
Alpha	Days 1–30	Internal team + invited users in Luang Prabang	20 tourist profiles; 10–15 guide profiles; 50 simulated bookings
Beta	Days 31–60	Invite-only cohort (~100 tourists)	NPS survey; guide acceptance rate; recommender re-evaluation
Beta II	Days 61–90	Open sign-up; 1–2 additional Laos destinations	Safety/trust score validation; XGBoost predictor wired if prototype validates

Launch Decision

Launch Question	Current Answer
Public launch now?	No, not as full production
Controlled pilot?	Yes, with selected operators/hotels in Luang Prabang
Validate during pilot?	Conversion, satisfaction, cancellation, guide acceptance, safety escalations
Main risk?	Synthetic data may not reflect real tourist-guide behaviour
Mitigation?	Pilot with real outcome tracking and human oversight

Final Recommendation

Wanderless Laos should **not** be launched as a full public production system yet. It should proceed to a **controlled 90-day pilot** with 3–5 selected local operators or boutique hotel partners in Luang Prabang. The product is technically credible: the hybrid recommender (cosine + TruncatedSVD) is a legitimate

dual-signal approach, the itinerary optimiser (greedy + 2-opt) is a proper two-stage metaheuristic, and the group formation module (K-Means + DBSCAN) provides structured segmentation. The safety/trust indicator is a decision-support signal — it helps tourists form a judgement, it does not certify outcomes.

Page 5 — AI/ML and Decision-Support Design

ML Architecture

Wanderless Laos uses five ML-adjacent modules. One is a full ML pipeline (recommender), three are algorithmic (group formation, itinerary, pricing), and one is a rule-based decision-support prototype (safety/trust score). An additional XGBoost prototype exists but is not wired to the live API.

Component	Method	Purpose
Preference representation	Normalized feature vectors (food/culture/adventure/pace/budget)	Compare tourist and guide attributes numerically
Guide matching	Cosine similarity between preference and expertise vectors	Rank guide compatibility
Collaborative filtering	TruncatedSVD (k=20) on tourist-guide rating matrix	Surface rating-pattern insights for sparse ratings
Segmentation	K-Means clustering + DBSCAN outlier detection	Identify traveller group patterns
Itinerary logic	Greedy construction + 2-opt local search	Sequence preference-aware daily routes
Safety/trust indicator	Rule-based weighted score (license, language, density, fit)	Support safety comfort judgment — not a guarantee
Dynamic pricing	Rule-based quote (daily rate × duration × season × group)	Provide simulated price estimates
Satisfaction prediction	XGBoost prototype on synthetic ratings	Prototype only — NOT wired to API

This is not only a BI dashboard. The app ranks tourist-guide-itinerary combinations and returns an **explainable recommendation** — each guide's score traces to component factors.

Explainability

- Each compatibility score is decomposed into: content-based fit, collaborative rating fit, and destination boost
- Top match factors and trade-offs are displayed for each guide recommendation
- The safety/trust score shows its component signals (license verification, language alignment, tour density)
- Limitations are disclosed: the score is a decision-support prototype, not a safety certification

Governance

- All training data is synthetic and labelled `rating_source: "synthetic"`
- Safety/trust score is a rule-based prototype — not a guarantee
- Human escalation path required for safety-related disputes during pilot
- Future pilot data must be handled under a privacy-safe framework
- New guides with few ratings may be disadvantaged by collaborative filtering — fair-exposure monitoring needed

Page 6 — Technical Architecture and Developer Handover

Repository Structure

Area	Path	Purpose
README	<code>README.md</code>	Main navigation and quick start
Backend	<code>backend/</code>	FastAPI API server, matching logic, ML modules
Frontend	<code>app/</code>	Flutter cross-platform mobile app
Data	<code>data/</code>	Synthetic data generator and output CSVs
Model docs	<code>MODEL_CARD.md</code>	Model logic, training data, limitations
Architecture	<code>AI_ML_ARCHITECTURE.md</code>	Pipeline and governance
CO config	<code>CO_CONFIGURATION.md</code>	Human-AI workflow design
Validation	<code>VALIDATION_REPORT.md</code>	Test results and smoke-test scenarios
Demo fallback	<code>DEMO_API_COMMANDS.md</code>	curl-based API demo if Flutter unavailable

Developer Flow: Request to Recommendation

```

Tourist enters preferences in Flutter app
→ Frontend sends GET /api/recommendations?
tourist_id=...&destination=Luang%20Prabang
→ Backend loads tourist profile from SQLite
→ HybridRecommender scores all Luang Prabang guides:
    45% cosine similarity (preference vs expertise)
    45% TruncatedSVD collaborative rating
    10% Luang Prabang destination boost
→ Optional: form_groups() segments tourists for group matching
→ Optional: build_itinerary() produces optimised route
→ Backend returns ranked guide list with scores and explanations
→ Frontend displays: guide cards, safety/trust indicator, simulated
booking

```

Running the Application

```
# Backend
cd backend
python -m pip install -r requirements.txt
python seed_accounts.py      # Seeds test accounts + Luang Prabang guides
python main.py               # Starts FastAPI on localhost:8000
# API docs: http://localhost:8000/docs

# Tests
cd backend && python -m pytest ../tests -x -v
# Expected: 14/14 tests passing

# Frontend (if Flutter is set up)
cd app
flutter pub get
flutter run
# Fallback if Flutter unavailable: DEMO_API_COMMANDS.md
```

Test Accounts (password: wanderless123)

Role	Email	Details
Tourist	test@wanderless.com	Alex Traveler; S\$5,000 simulated wallet
Guide	guide@wanderless.com	Bounmy Phommasak (Luang Prabang specialist)
Business	business@wanderless.com	Luang Prabang Heritage Tours

Page 7 — Validation, Limitations, Risks, and Next Steps

Validation

Area	Status	Evidence
Backend tests	14/14 passing	VALIDATION_REPORT.md
Business smoke tests	5/5 scenarios documented	VALIDATION_REPORT.md
API fallback demo	Available	DEMO_API_COMMANDS.md
Real user pilot	Not yet conducted	Required before production
Payment integration	Not implemented	Booking is simulated

Limitations

All ratings, tourist profiles, and guide profiles are **synthetically generated** (88% signal + 12% calibrated noise) and labelled `rating_source: "synthetic"`. Real tourist behaviour may differ significantly. Safety/trust scoring is rule-based — no real-time GPS, incident reports, or third-party safety API integration. The XGBoost satisfaction predictor is a prototype not wired to any API. SQLite is appropriate for demo; PostgreSQL is required for production scale.

Risk Table

Risk	Impact	Mitigation
Synthetic data results not generalising	MEDIUM	Pilot with real outcome tracking
Hardcoded matching weights biasing recommendations	MEDIUM	Retune after pilot data collected
Safety/trust score misread as a guarantee	HIGH	Clear "decision-support prototype only" labelling in UI and docs
Demo setup variability (Flutter)	LOW	API fallback via <code>DEMO_API_COMMANDS.md</code> always available

Next Steps

Business: Secure 3–5 boutique hotel or local operator partners in Luang Prabang; onboard 10–30 guides; define pilot success metrics (guide acceptance rate, NPS, cancellation rate, safety escalations); establish human escalation process.

User validation: Conduct test sessions with real tourists; compare AI-guided match quality against manual recommendation; collect satisfaction labels, conversion, and cancellation data.

Developer: Confirm reproducible setup; harden API error handling; replace synthetic data with real pilot data; tune matching weights from outcomes; wire XGBoost predictor after sufficient real ratings gathered; add monitoring dashboard for guide exposure and fairness metrics.

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